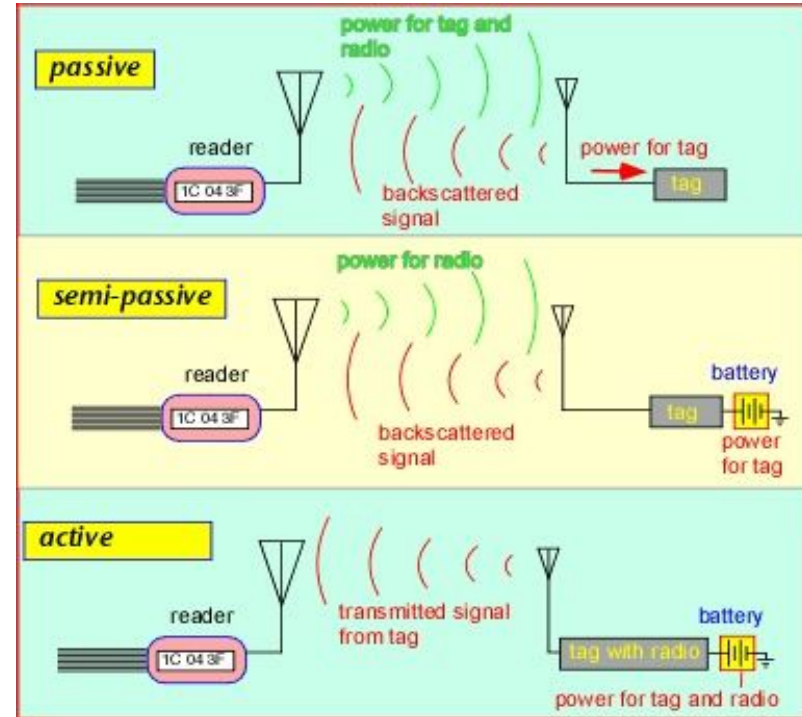


Mixed Signal NFC Temperature Sensor

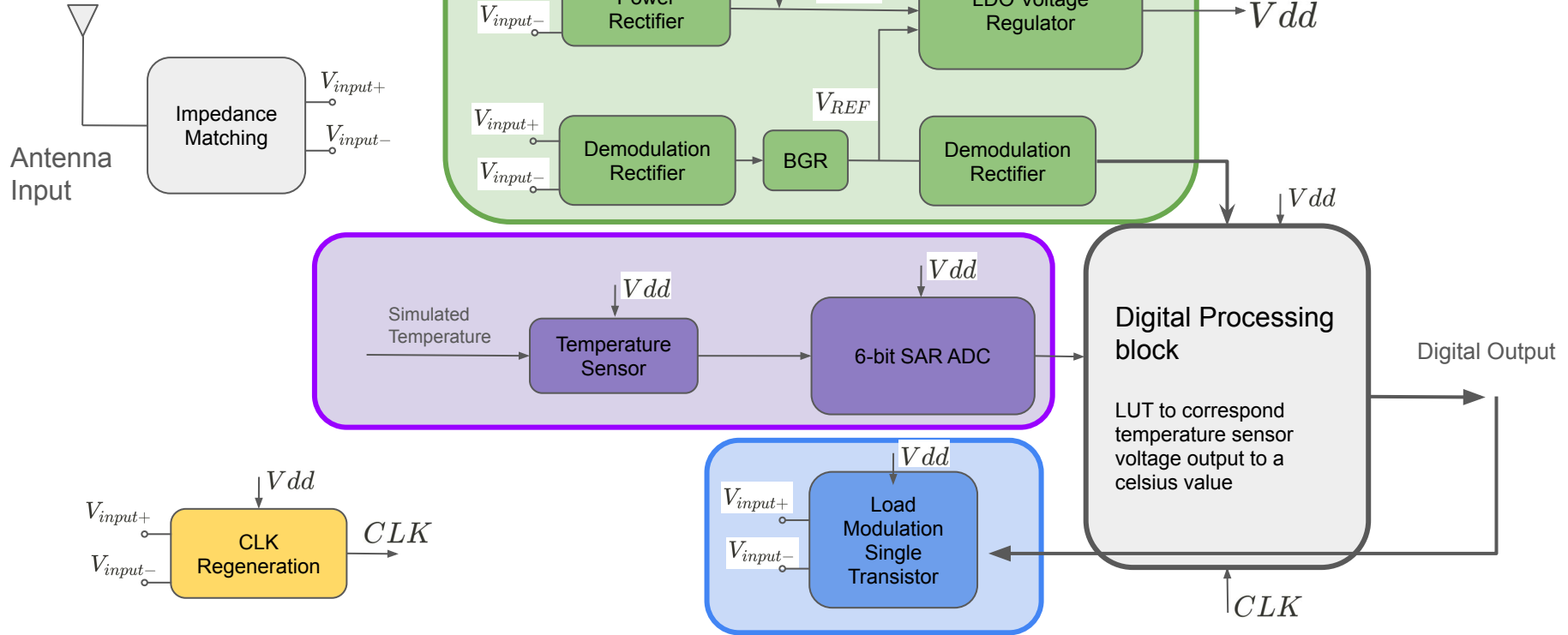
Giovanni Michel, Jack Thoene, Yuqing Wang

Project Background: Semi-Passive NFC

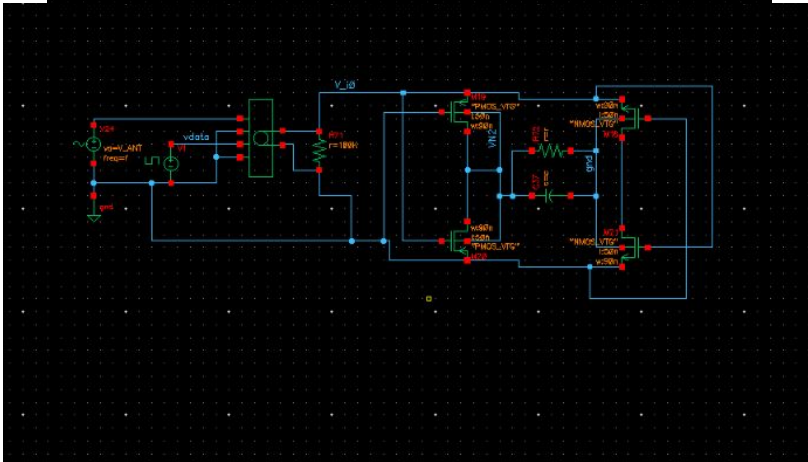
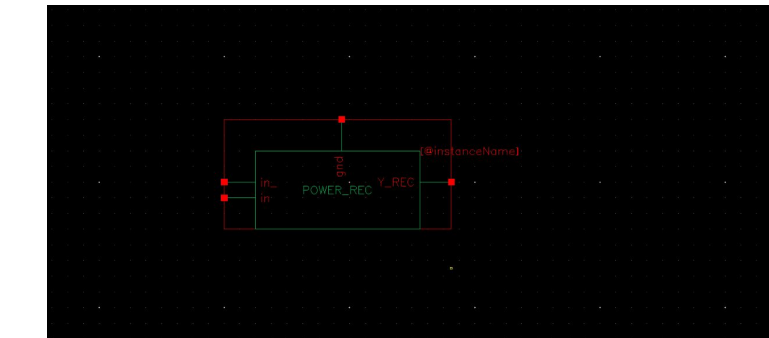
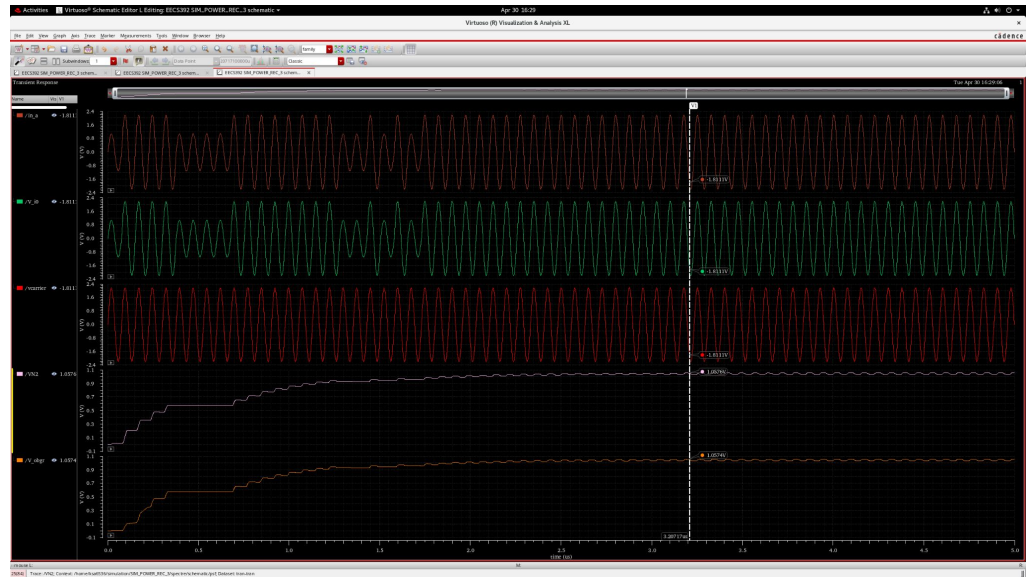
- Send temperature data using RF backscatter
 - Specific frequency 13.56MHz, applicable for short distances (~20cm)
 - e.g. secure contactless payment, key cards, etc
- Power harvesting through inductive coupling
- Small battery to enable logic on chip



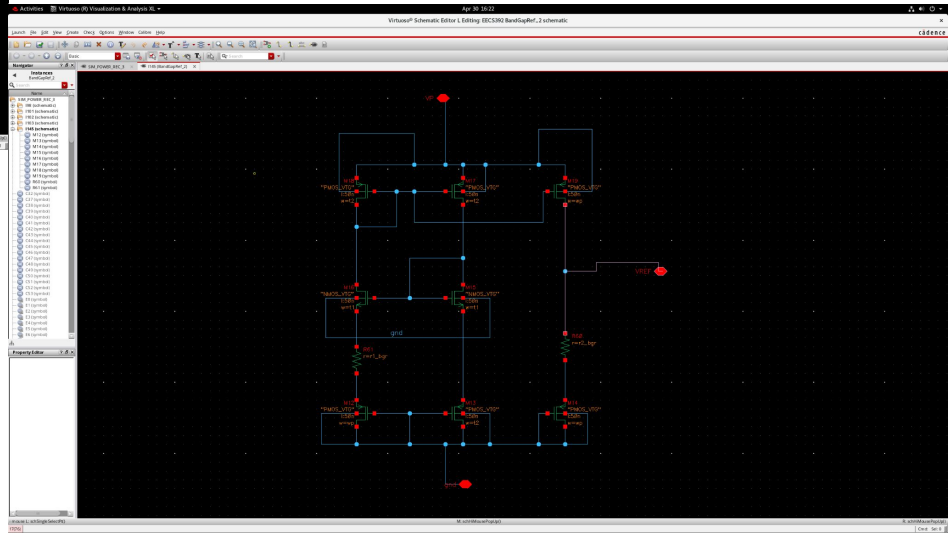
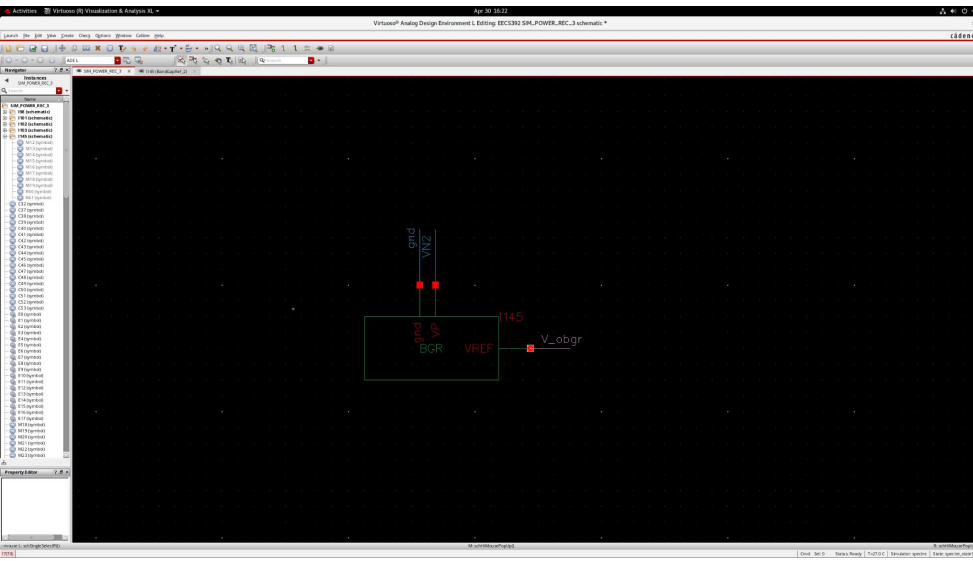
Circuit Block Diagram



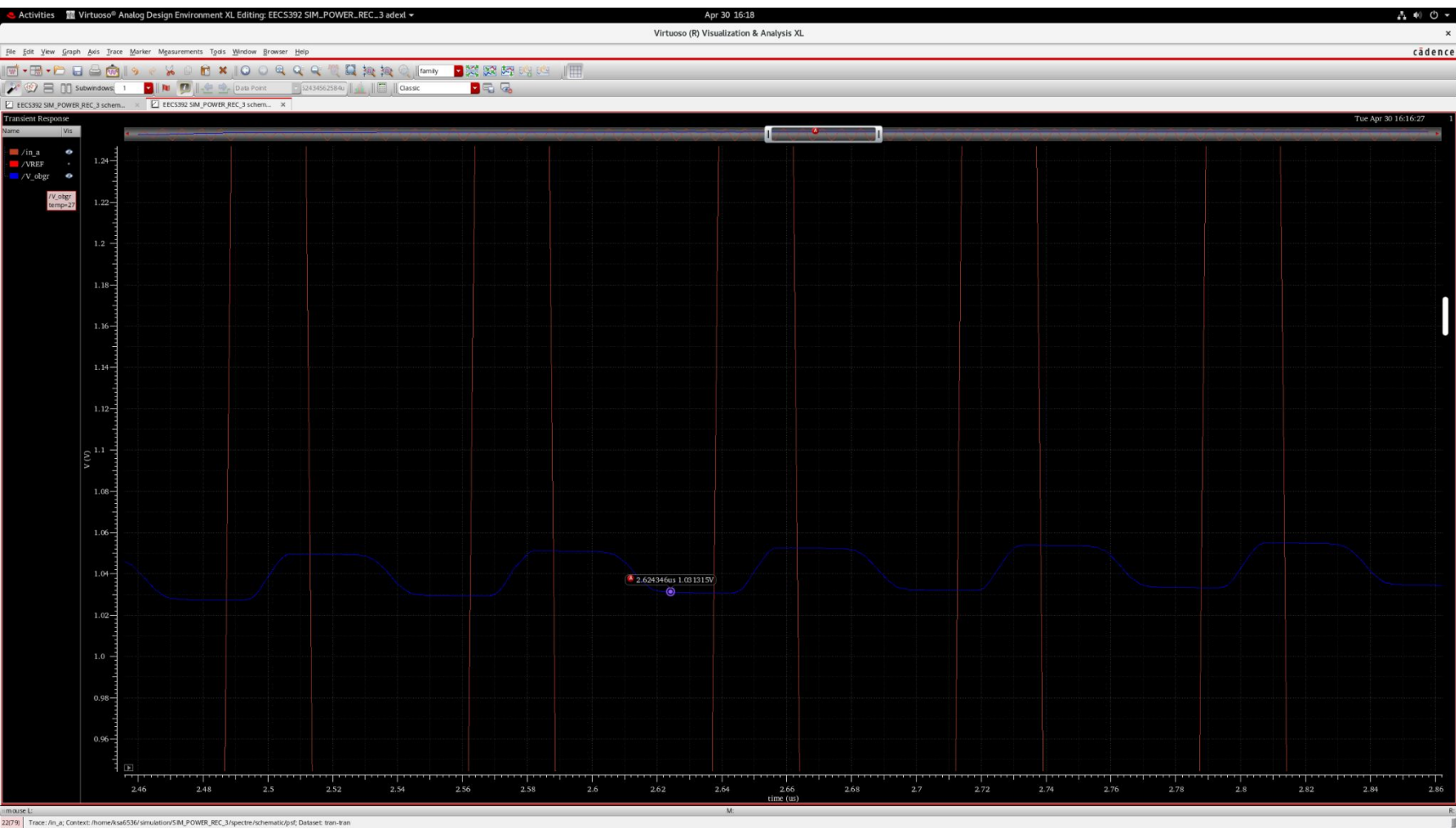
Power Rectifier



Bandgap Ref circuit



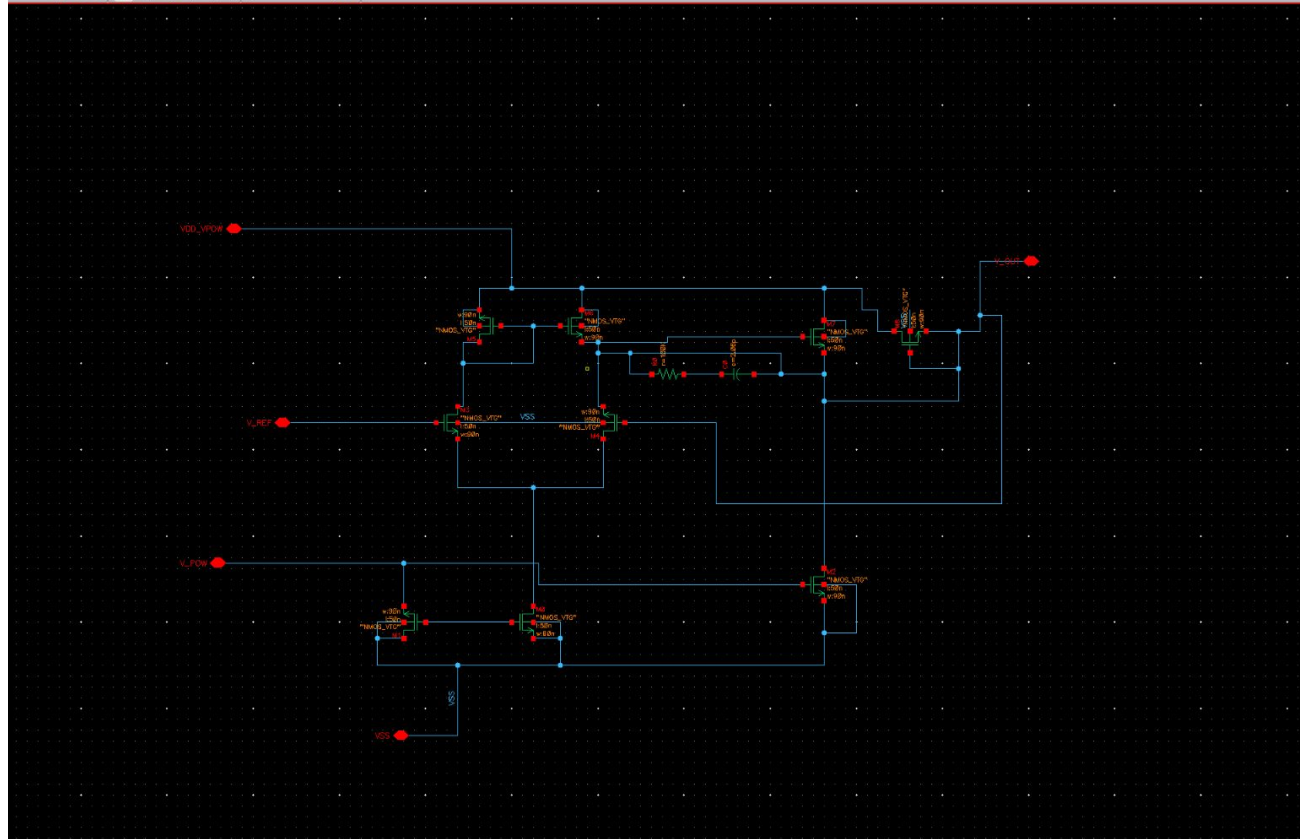
Bandgap Ref circuit @27C



Bandgap Ref circuit @-27C



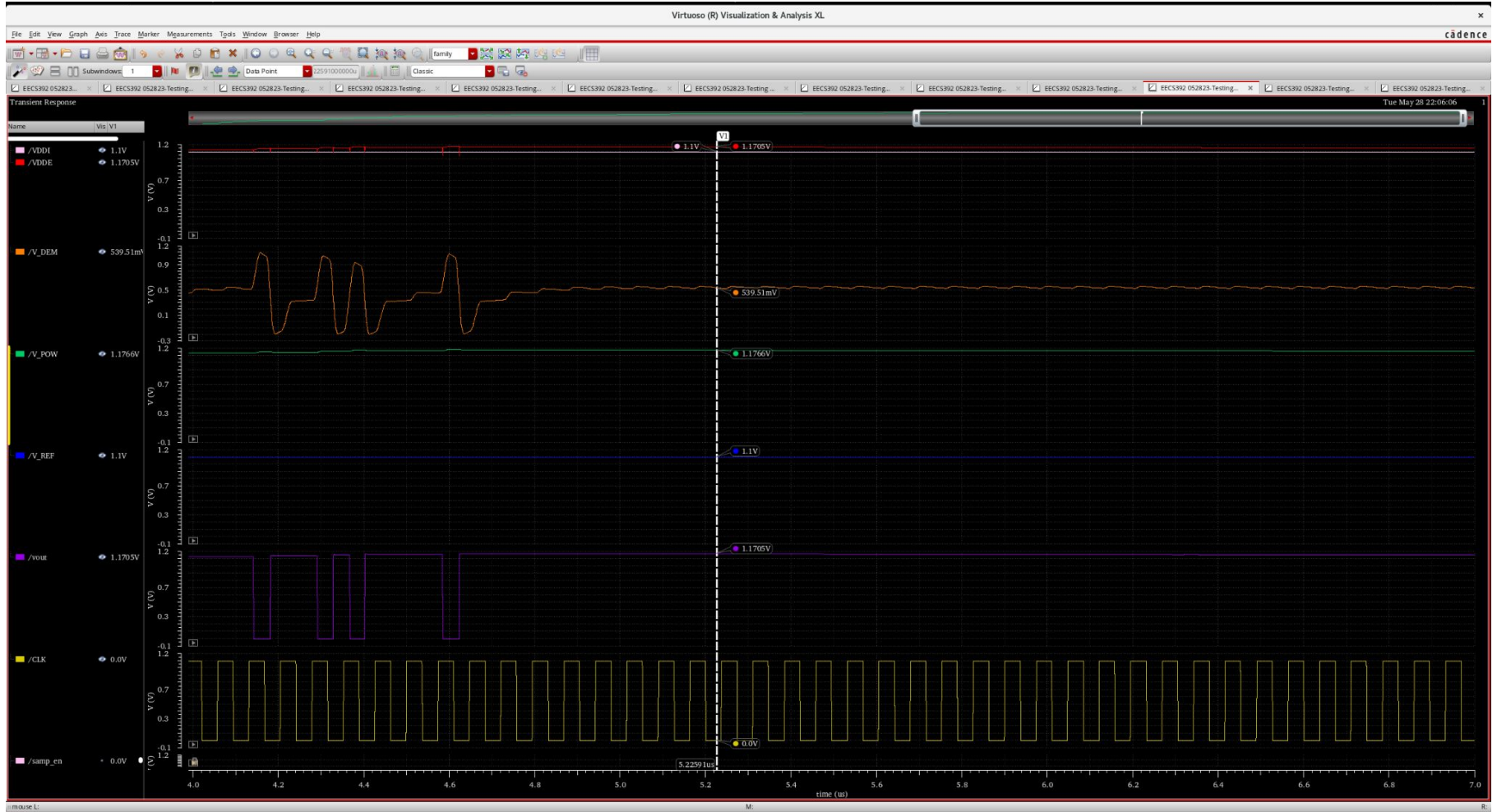
Error amplifier (LDO)



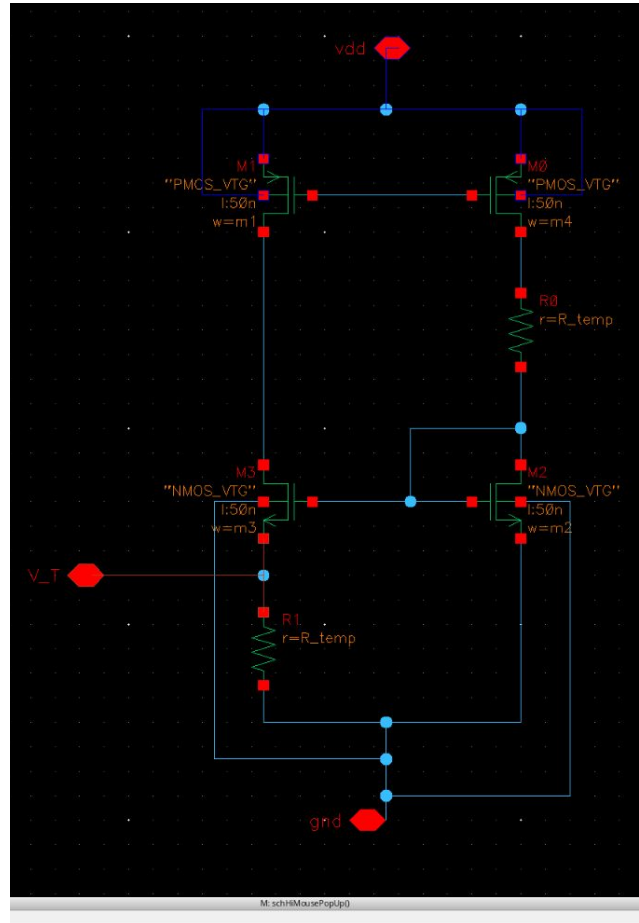
LDO Voltage Regulator



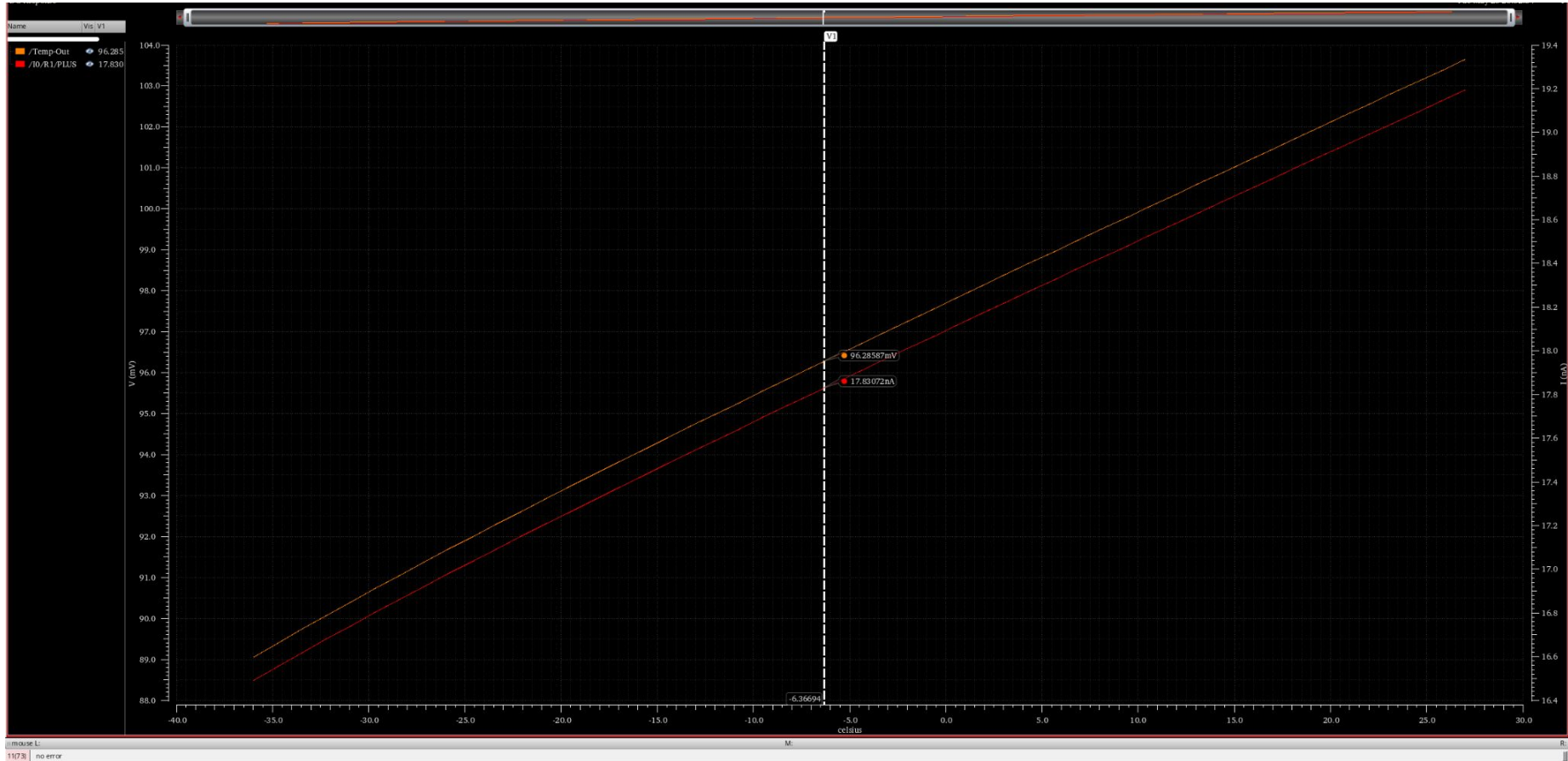
Power Components



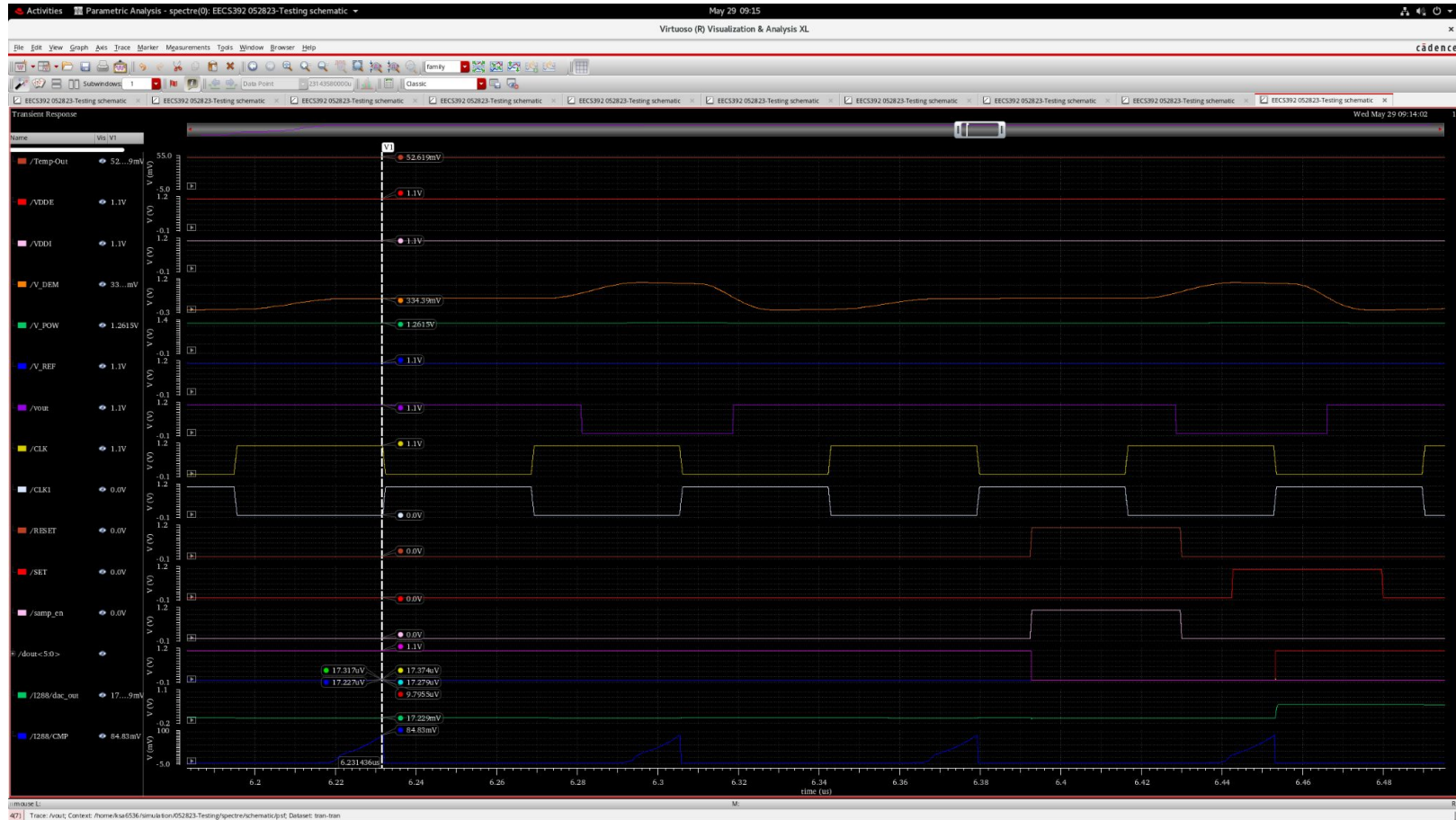
Temperature Sensor Schematic



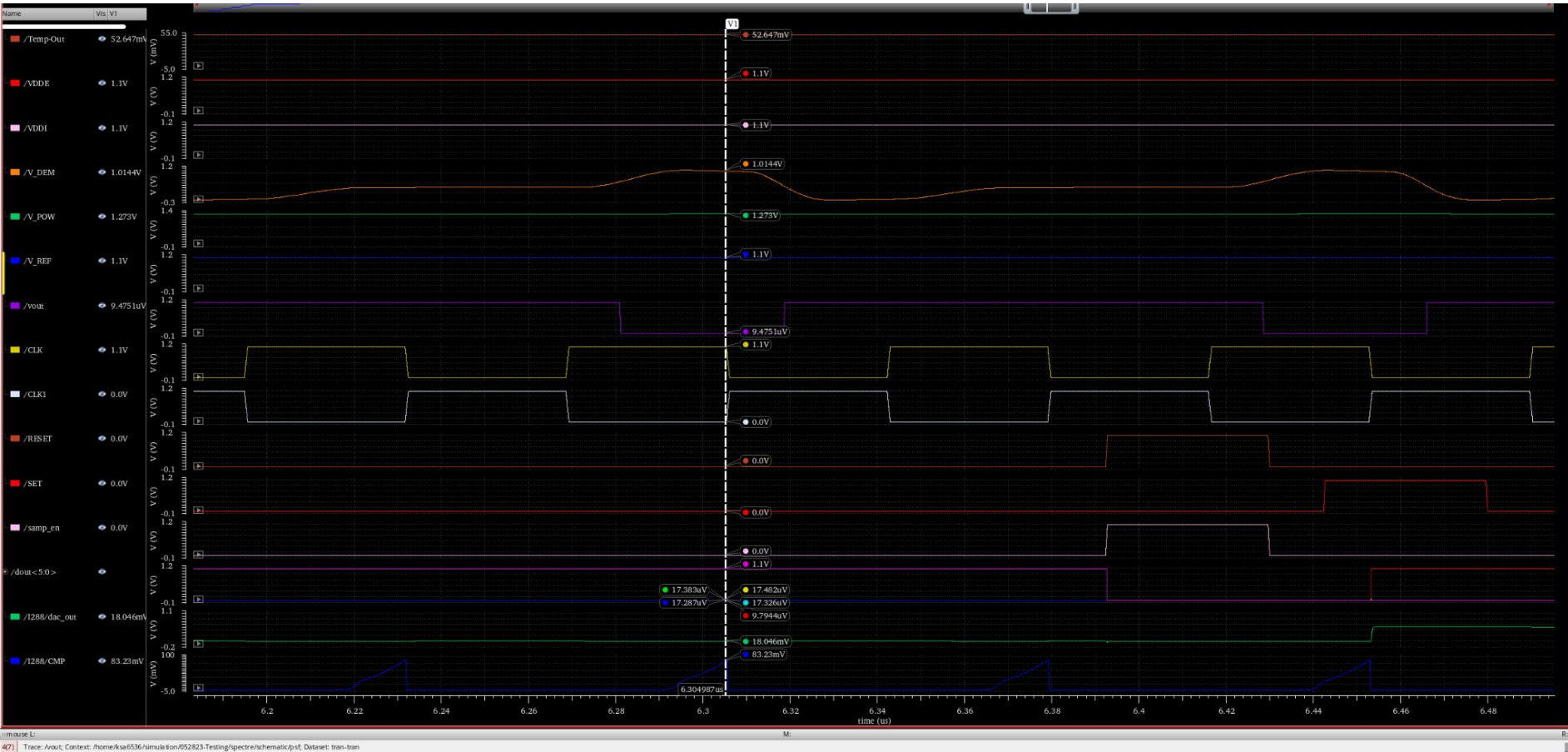
Temperature Sensor Simulation



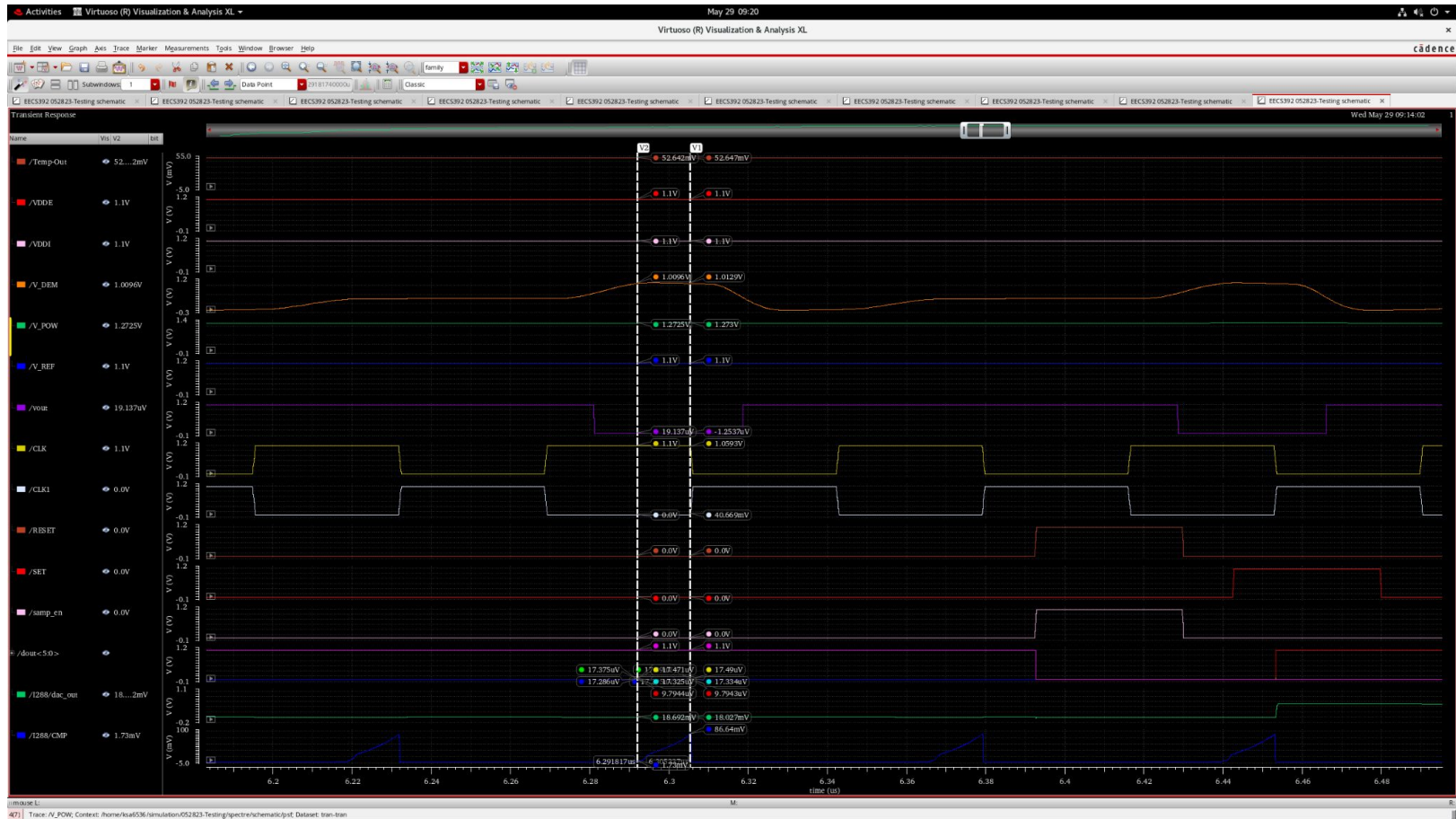
Temperature Sensor and Power Components with ADC/DAC



Temperature Sensor and Power Components with ADC

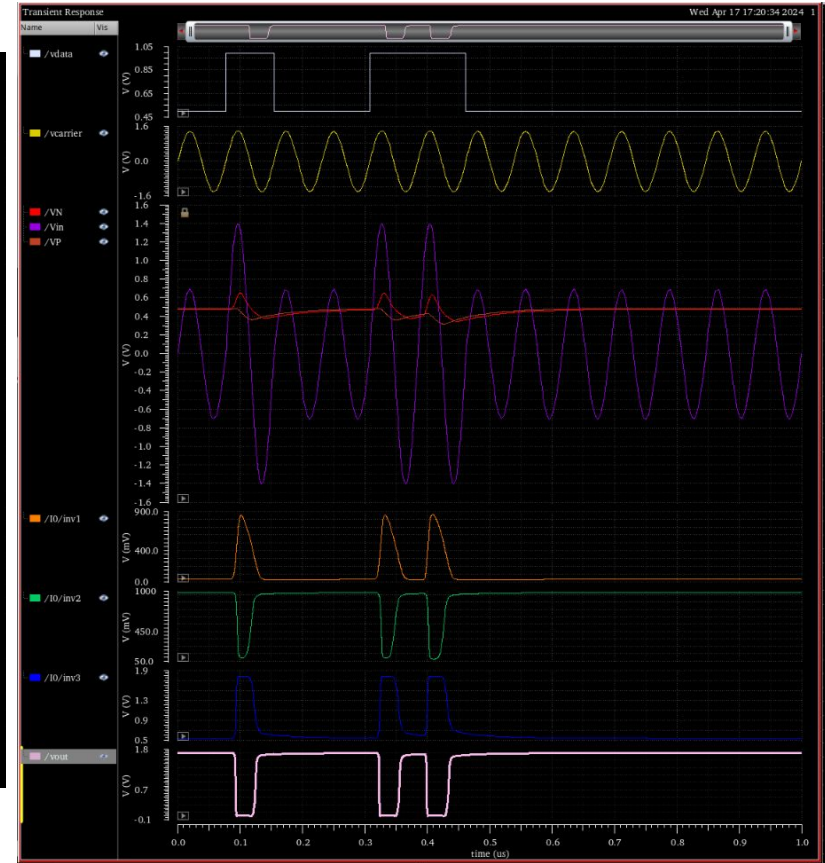


Temperature Sensor and Power Components with ADC/DAC



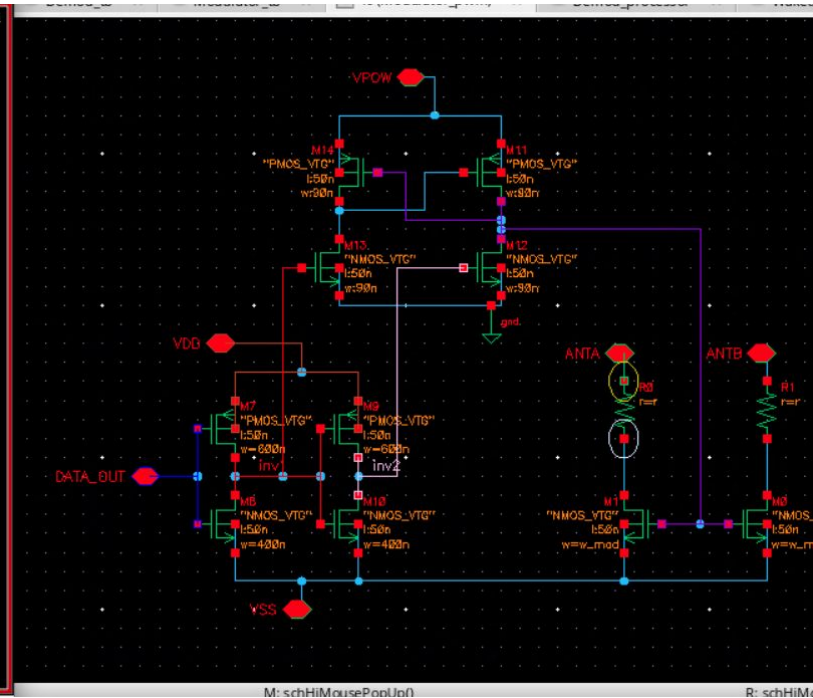
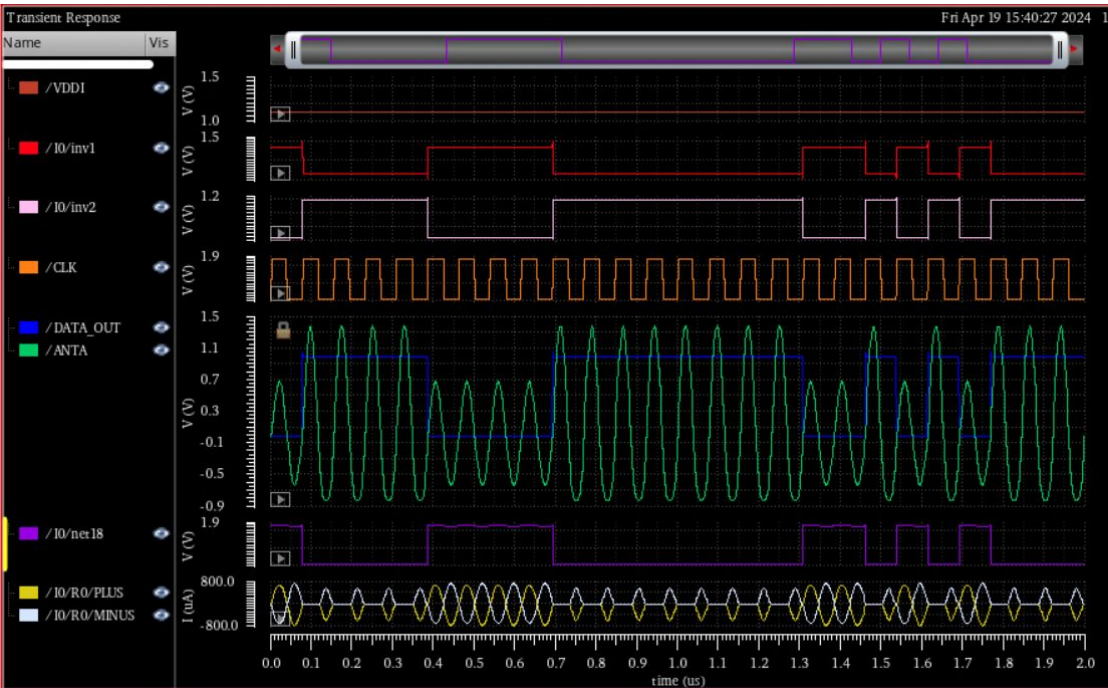
Demodulation & Output Modulation

Input Demodulation



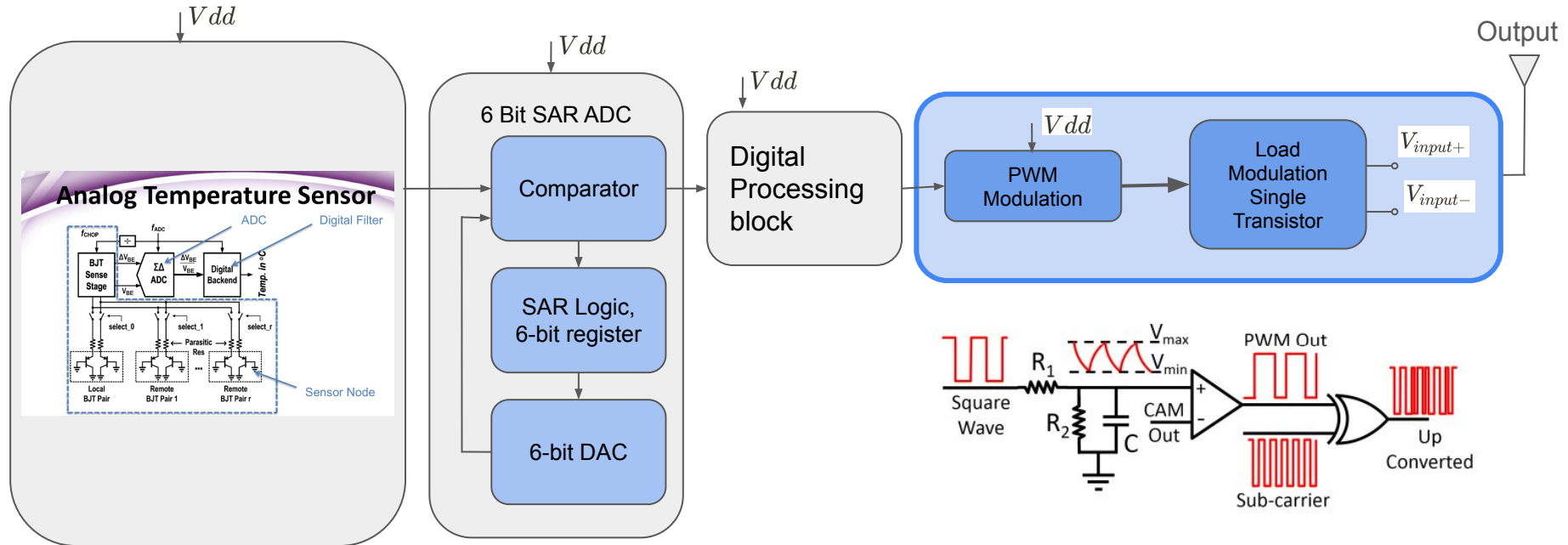
Output modulation

Takes output from digital block(blue) and modulates the antenna voltage (green)

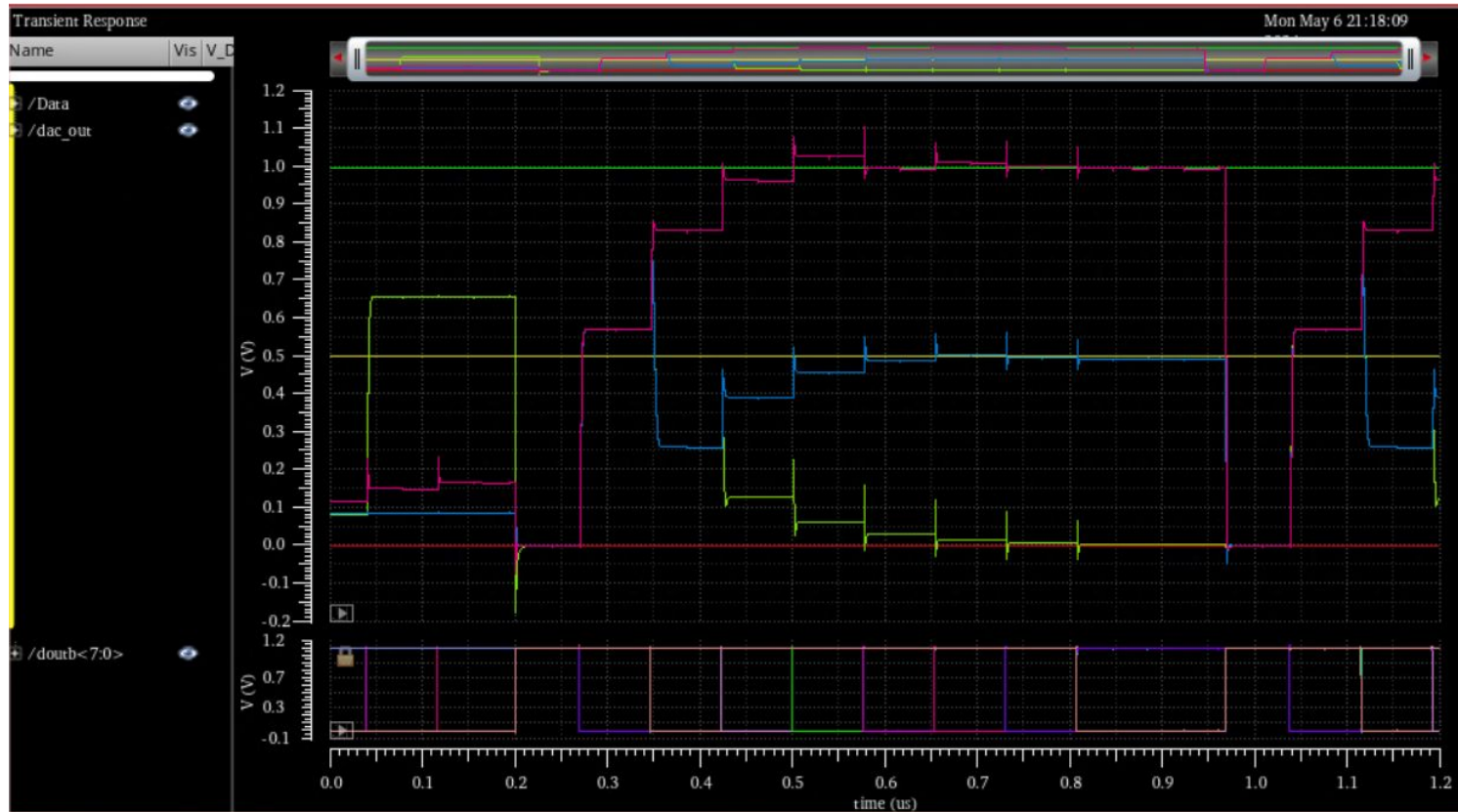


Sensor datapath

- Low power consumption Successive Approximation Register ADC
- Digital (ASK/PSK) and analog PWM modulation on output signal



SAR ADC

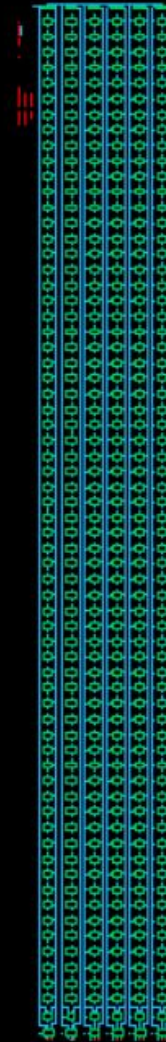
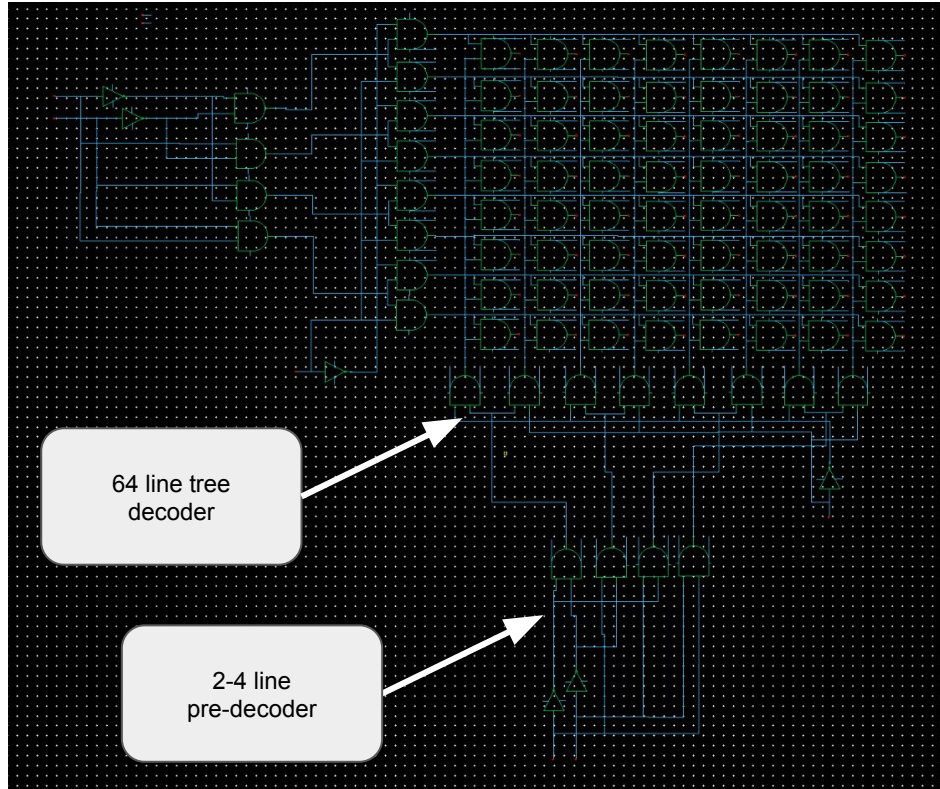


SAR ADC -> Register shifted output

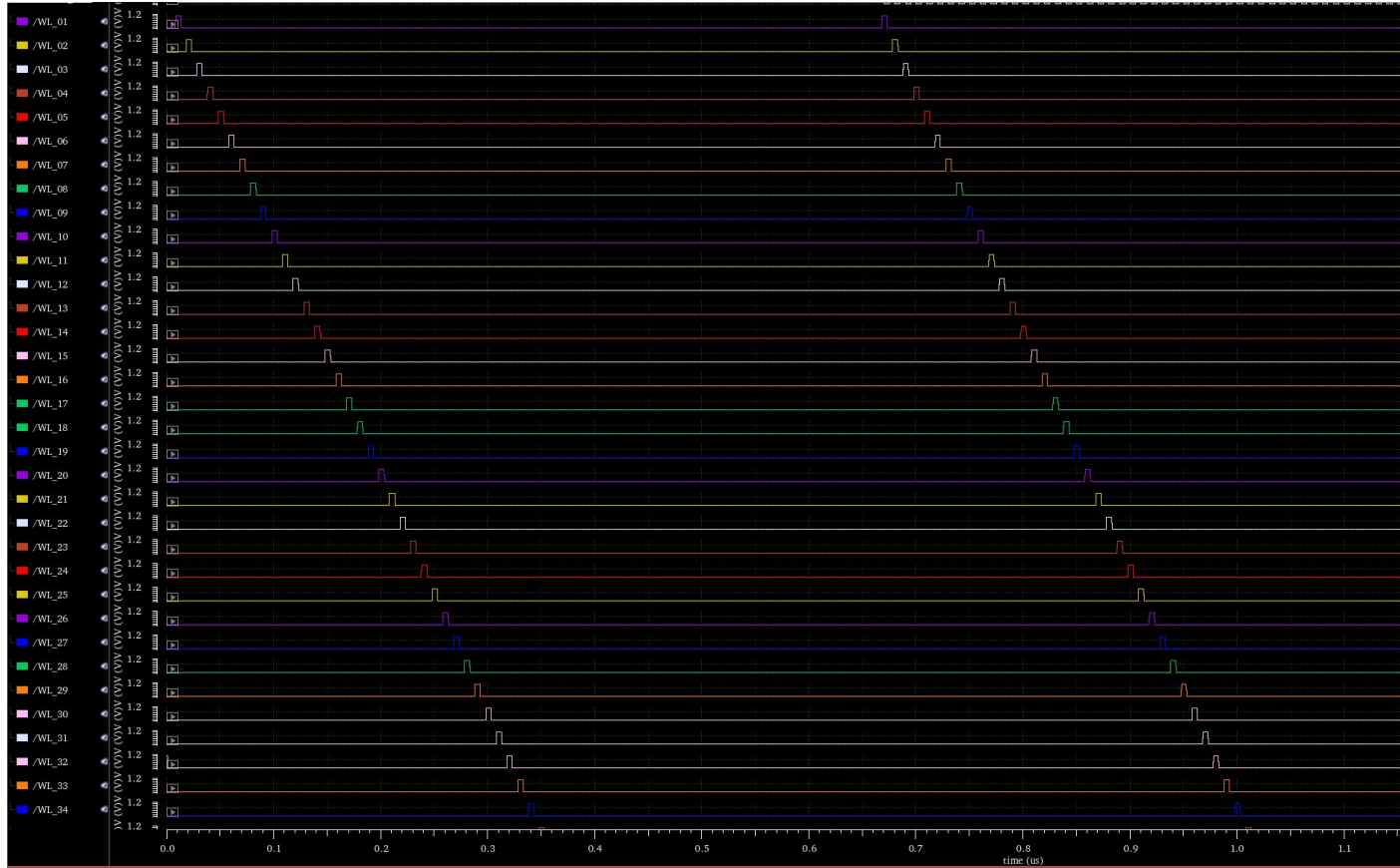


6-64 Decoder & SRAM

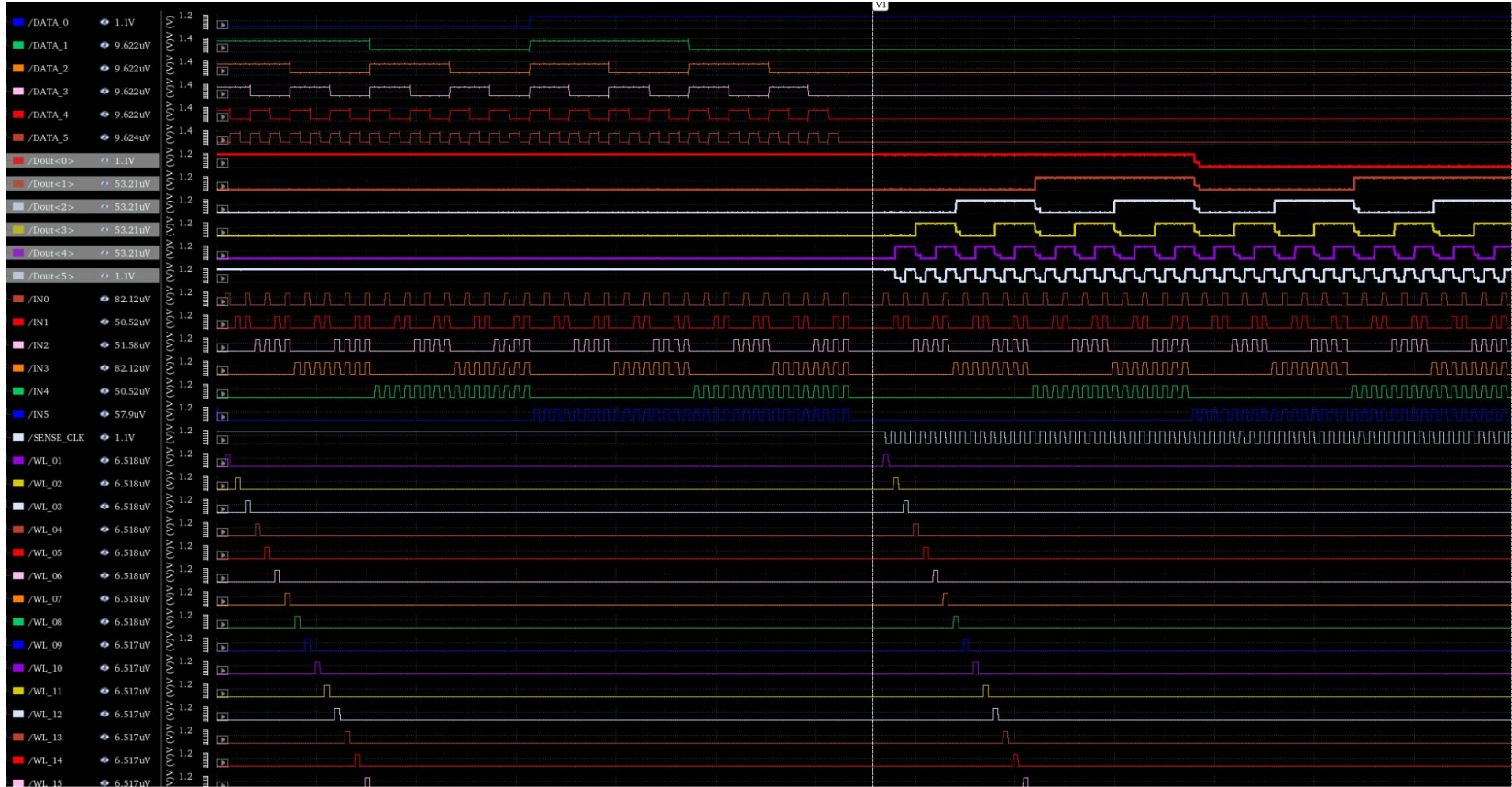
6-64 Decoder $\rightarrow$ 6x64 SRAM



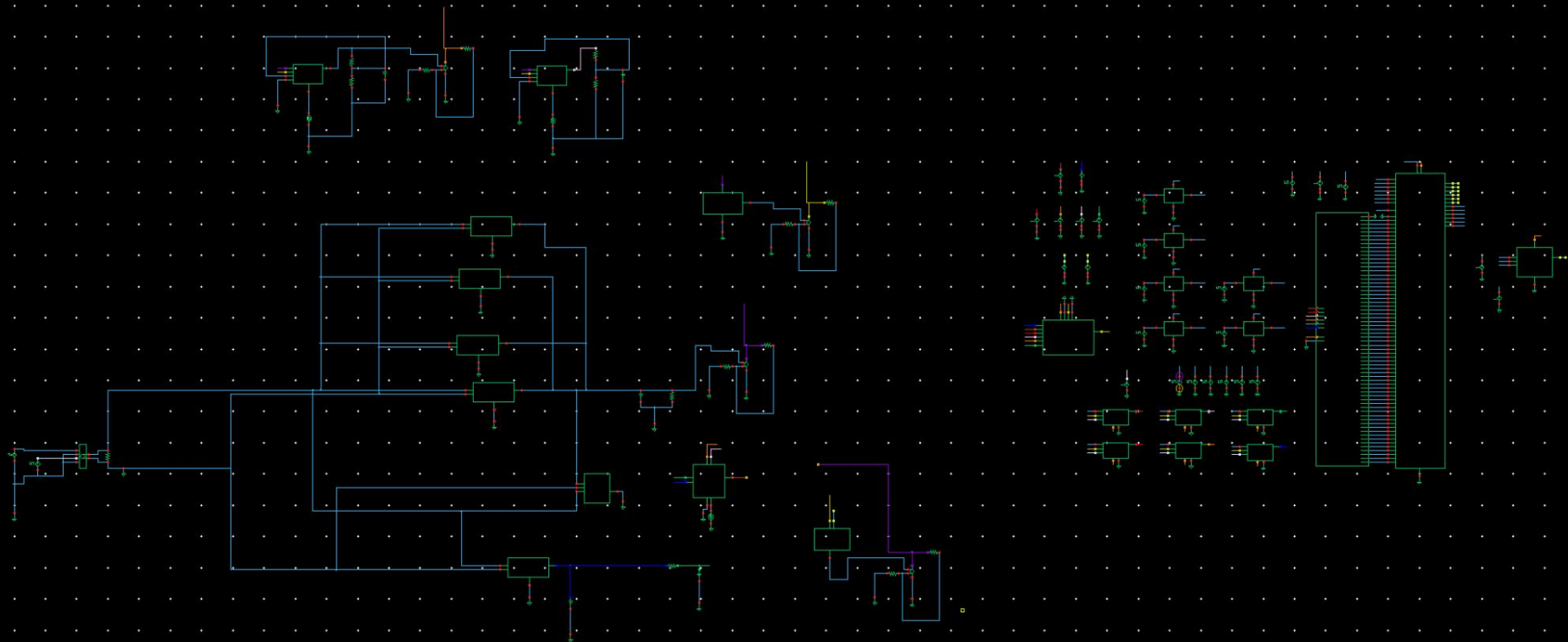
6-64 Mux Output Simulation



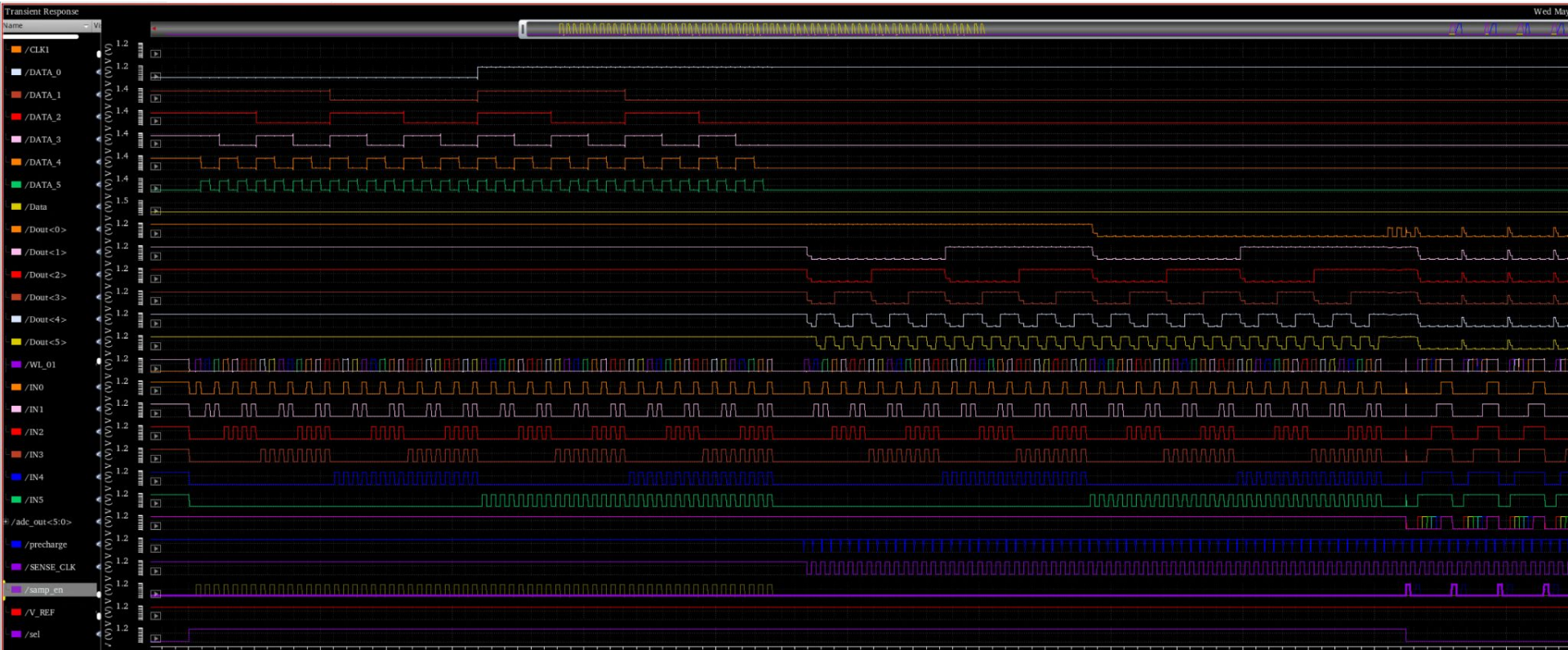
Digital LUT simulation: Read & Write



Combined Testbench



Combined Testbench



Results & Other considerations

- Overall deliverable completed: ASIC Chip Simulation in Cadence Virtuoso (analog components as well as the schematic equivalent of the digital LUT)
 - Input: Temperature data for corner analysis, RF signal voltage and resistance
 - Output: data sent through RF (shown through modulation of the antenna voltage) of a binary value representing the temperature sensor reading in celsius
- Ethicality: disposable asic environment risk, security
- Future: self-generated or RX programming capabilities, further reducing power

<u>Giovanni</u>	<u>Yuqing</u>	<u>Jack</u>
Power Rectifier	Demodulation RX & Load modulation TX	Digital LUT
LDO, BGR	SAR-ADC	SRAM integration
Temp Sensor, SRAM	SRAM & shift register	Digital Filter